# 2026R1 Translation Technology in the Age of AI Final Project: Build a Translation and Linguistics Corpus Explorer

Course: Translation Technology in the Age of AI (2026R1)
香港中文大學翻譯系

Due 5 December 2026

**Individual assignment.** Your deliverable is a **website**, published on GitHub Pages, that contains an interactive corpus explorer and a learning report.

> Submission deadline: **5 December 2026**
> Submit: the URL of your GitHub Pages site (and the URL of your GitHub repository).

## What you will do

Pick a topic related to translation or linguistics that interests you. Build a small corpus of **500 entries**, then use an AI agent to help you build a website where visitors can **search and filter** the corpus and see it through **at least three interactive visualizations**. On the same website, write a **learning report** that tells the story of your project.

You do not need to write code by hand. The point is to learn how to **direct an AI agent**, **check its work**, and **think critically about translation data**.

## Choosing a topic

You choose your own topic and title. Ask yourself:

1. Can I get 500 entries from an existing, traceable source?
2. Is there a visible translation pattern, strategy, or disagreement I can classify?
3. Can I explain my findings with a concept from translation studies?
4. Will search, filters, and charts actually help my analysis?
5. Does my topic involve profanity, politically sensitive material, or personal data? If so, how will I handle it?

Appendix A lists topic ideas, and Appendix B gives a topic checklist. Short entries (signs, menu items, station names, terms, headlines, short subtitle lines) are easier to collect and check. An entry does not have to be a single word: a sentence pair, or a key term you extract from a sentence pair, also counts, as long as you record the unit in the `unit` field.

## The corpus

- **Size:** at least **500 entries**.
- **Use existing corpora.** Please start from a ready-made source rather than typing data by hand. Examples to explore (always check each source's terms of use):
  - **OPUS** (opus.nlpl.eu): large collection of parallel corpora, including **OpenSubtitles**
  - **OpenSubtitles 2024** parallel subtitles (English–Simplified Chinese, `en-zh_CN`). This is the corpus used in the course example site, so you can compare your work with it. The corpus is large (about 22 million sentence pairs), so you must **sample** from it (see below).
  - **Taiwan's National Academy for Educational Research (NAER, 國家教育研究院)**: its terminology database, 樂詞網 (https://terms.naer.edu.tw/). This is mainly word-level terminology, so it suits term-level entries.
  - **HK Open Data Portal** (https://data.gov.hk/): open datasets from Hong Kong government departments, including bilingual lists such as place names, public facilities, and public notices. Check the licence of each dataset (the portal's terms allow free use with clear attribution to the source and to DATA.GOV.HK).
  - Hong Kong government bilingual materials (e.g., laws, press releases, place and street names)
  - **Hong Kong e-Legislation** (https://www.elegislation.gov.hk/): the electronic version of Hong Kong ordinances, in English and Chinese. Its contents are protected by copyright, so check the Copyright Policy (https://www.elegislation.gov.hk/copyright) and the Terms of Use (https://www.elegislation.gov.hk/terms) before use. Keep to short quotations with the section cited, and record the URL and retrieval date in `source`.
  - Your own collection of public signs, menus, or app screens (with photos or screenshots)
  - **Tatoeba** (https://tatoeba.org/): downloadable sentence pairs in many languages, under its own license
  - **Bilingual data collected with a crawler.** You may write a small crawler to collect bilingual pages, for example bilingual Hong Kong government press releases or the bilingual texts on the Hong Kong e-Legislation site. Check `robots.txt` and the site's terms first, rate-limit your requests, record the page URL and retrieval date in `source` for every entry, and keep only the sentence pairs you need, not whole documents.
- **Sample, do not cherry-pick.** Taking the first 500 lines, a block of 500 consecutive lines, or the entries you like best will not give a fair picture of the corpus. Use a documented, reproducible method, for example:
  1. Draw random sentence IDs with a fixed random seed, and write the seed down.
  2. Keep a pair only if it passes filters you state in advance (for example: both sides non-empty, English contains letters, length within a range).

3. Record the method, the seed, the filters, and the number of candidates rejected in your report's Method section.

- **Use an AI agent to collect and clean the data** (extract, deduplicate, format, normalize).
- **Never invent data.** Do not ask AI to generate sentences, translations, or sources that do not exist. Fabricated data will be penalized and must be redone.

### 0.3.1 Required fields

| Field | Description |
|---|---|
| `source_text` | The original text |
| `target_text` | The translation |
| `source` | Where the entry comes from: corpus name and version, the sentence ID or URL, and the date you collected it (for example: `OpenSubtitles 2024 en-zh_CN, sentence id 7180, sampled 2026-11-02, seed 2026`) |
| `category` | Your classification (e.g., translation strategy, domain, text type) |
| `human_checked` | `yes` or `no`: whether you personally verified this entry |

Optional fields (metadata): `unit` (term, short phrase, or sentence pair), `key_term` (the word or phrase you extracted for analysis, if any), year, genre, translator, region, notes, and so on. If you use `unit` or `key_term` in your analysis, include them in your data file.

### 0.3.2 Quality check

Manually verify at least **50 entries** against the original source, and make sure every entry you check is correct. Mark them `human_checked = yes`. Sample them randomly, state how you sampled them in your report, and report how many errors you found and corrected (for example, "I checked 50 entries and found 4 errors, which I corrected"). Perfect data is not required. A system that works, with honestly reported limitations, is what matters.

### 0.3.3 Codebook (required)

Before you code your 500 entries, write a **codebook**: a short document that defines each category, gives rules for applying it, and shows examples.

For each category, include:

1. **Name**
2. **Definition** (one or two sentences)

3. **Theory source** (for example, Venuti 1995 on domestication, or Vinay & Darbelnet on translation procedures)
4. **Decision rule** (how you decide the category for an entry)
5. **Positive example** and **negative example**
6. **Boundary case** (an entry that could fit two categories, and how you decided)

Use three to six categories. Test them on 20 to 30 entries first, then revise. Keep a short log of every change to your codebook and why you made it. Put this log in the Method section of your report, next to your manual check of the 50 entries, because together they show how reliable your data is.

AI can draft the codebook and apply it to your data. However, **you must rewrite the definitions in your own words**, and you must be able to explain each category in your oral presentation. The example codebook is at `codebook.html` on the example site (it is an illustration, not a model answer).

## Website requirements

The website must be **static** (HTML, CSS, JavaScript, and a data file such as JSON). No server or database is needed.

1. **Search**: visitors can search the corpus by keyword.
2. **Filters**: visitors can filter by at least `category` and `human_checked`.
3. **Visualizations**: **at least three** interactive visualizations. Simple bar charts and pie charts alone do not meet this requirement. Choose at least **three different visualization types** from the families below, and make them fit your research question:

    - **Relational:** interactive network graph (words and their translations), chord diagram (flows between English and Chinese terms), Sankey diagram (volume flowing from source to target categories)
    - **Hierarchical:** treemap or sunburst (breakdown of categories into sub-categories)
    - **Lexical:** word cloud, or a bilingual KWIC (key word in context) viewer
    - **Distributional:** heat map (for example, category by source), scatter plot, or timeline
    - **Tabular:** an interactive data grid with sorting, filtering, and CSV export
    - **The visualizations must respond to the filters**: when a visitor changes a filter (`category`, `human_checked`, and any other field you use), every visualization and the data grid update.
    - **The visualizations must be linked to each other and to search.** Clicking an element in one visualization (a node, a word, a slice, a band in a Sankey) must update the other visualizations and the table. A word or term shown in any visualization must be clickable and open the **search page** with that term already searched. Search state (filters, selected term) must be in the URL, so a visitor can share the exact view.

- **Give at least one visualization real controls**, not only a click. For a network graph, examples are: number of top terms, minimum link weight, keyword filter, zoom and pan, dragging and pinning nodes. A visitor should be able to change the view without editing code.
- Each visualization needs a **caption** stating the question it answers (one sentence). Your report's Analysis section must refer to each visualization by name.
- Check your numbers. Each chart must match the underlying table for the same filters.

4. **Learning report** (a page on the same website), with text and images/charts. Include these sections:
    1. **Origin**: why you chose this topic
    2. **Background**: what led you to it
    3. **Gap**: what is missing or unclear in what already exists
    4. **Questions**: what problems you tried to solve
    5. **Method**: how you built the corpus and system
    6. **Analysis**: what the data shows (use your charts)
    7. **Conclusion**: what you achieved
    8. **Reflection**: what you learned, what was difficult, what you would do differently, and what AI could and could not do for you
5. **AI Use Declaration** (a short page or section): see Section 6.
6. **Codebook page**: your codebook (see Section 3) must be published on the site.
7. **Bilingual interface**: the site must be switchable between **English and Traditional Chinese (繁體中文)**. The default language is English. You may keep the data in its original language; only the interface text (headings, labels, buttons, report text, codebook text) must be bilingual. A simple language switch (like the one in the example site) is enough.
    - Do not translate your corpus data unless you want to. Your Chinese report text should be in Traditional Chinese.

A working example site is at https://nova-abbott.github.io/tt-example-site/ (source code in the repository `tt-example-site`). It uses 500 OpenSubtitles 2024 sentence pairs and has six visualizations: a bar chart for each of three fields, a co-occurrence network, a Sankey diagram, and a treemap. Use it as a reference for how the pieces fit together, not as a model to copy: your topic, categories, and visualizations must be your own.

Our department's **NV Atlas** (https://nvatlas.tra.cuhk.edu.hk/) is a larger, finished project you can look at for inspiration. It has three search modes, six visualization interfaces (network graph, treemap, word cloud, Sankey, sunburst, and chord diagram), and About, Statistics, and Feedback pages. Another department site, **WTF-Trans** (https://wtf-trans.tra.cuhk.edu.hk/), applies the same concept to a different topic (translation of foul words) and is still under development. Its network graph has finer controls (for example:

how many top terms to show, how many translations per term, a minimum link weight, keyword filtering, and pinning nodes); look at it for ideas. Be aware that it contains profanity. Look at how the visualizations on both sites link to each other and to search. Your site does not need to be that big, but at least three of its visualizations should reach that level of interaction.

## Publishing with GitHub Pages

1. Create a **public** repository in your GitHub account.
2. Put your site files (starting with `index.html`) in the repository.
3. In the repository, go to **Settings** > **Pages**, choose the `main` branch and the root folder, then save.
4. After a few minutes your site is available at `https://<your-username>.github.io/<repository-name`

Note: GitHub Pages sites are **public on the internet**. Treat your site as a portfolio piece you can show to future employers or programs.

## AI Use Declaration (one page)

Be honest and brief. Include:

- Which AI tools you used, and the underlying LLM (model name and version, if you know it)
- What you asked AI to do (e.g., collect data, clean data, write code, draft text)
- What you did yourself and what you checked by hand
- One or two example prompts that worked well
- One thing AI got wrong, and how you noticed

A full chat log is **not** required.

## Copyright and privacy (please read)

- Use only material you may legally use for coursework and public display. If you are unsure, check the source's terms of use. If you still cannot tell, choose different material.
- Do **not** put full copyrighted texts on your site (whole novels, whole episodes of subtitles, full song lyrics). Short quotations with a clear source are fine.
- Do **not** publish personal information (phone number, home address, or other people's private data). Your student ID may be published.
- Mark profanity or sensitive content clearly, and consider hiding it behind a warning.
- Your site is public. You are responsible for what you publish.

## Grading

| Area | Weight |
|---|---|
| Learning report: origin, gap, conclusion, reflection | 25% |
| Corpus quality: traceable sources, spot checks, consistent categories | 20% |
| Website: search, filters, interactive visualizations, codebook, bilingual interface | 25% |
| Analysis and conclusions (including codebook reasoning) | 20% |
| AI Use Declaration and reflection | 10% |

**Total: 100%.**
Your report on the website is the main written submission.

### 0.8.1 Oral presentation (graded separately)

- **Format:** in person, in class.
- **Length: five minutes** per student, followed by a short question-and-answer session.
- **Content:** explain your research: the origin, the gap, your main findings, and one example of how you classified an entry using your codebook.
- **Assessment:** the oral presentation is graded separately and is **not** included in the 100% above.

## Tips

- Start small: test your categories on 20 to 30 entries, then expand to 500 or more.
- Decide your categories **before** you collect everything, and write down what each category means.
- Save your work often (commit to GitHub).
- If something breaks, tell the AI exactly what you see (copy the error message or describe the screen).
- Check numbers in your charts against the table. AI can make charts that look right and are wrong.

Good luck, and have fun!

## Appendix A. Topic ideas

These ideas are starting points, not requirements. You may adapt one or choose your own. Check that your corpus source meets the requirements in Section 3 before you commit to a topic.

**Local linguistic landscape**

1. Chinese–English translations of shop and brand names in Hong Kong, Taiwan, and mainland China.
2. English labels for hospital and clinic signs (for example, 急症室, 門診): consistency across institutions.
3. Bilingual announcements and signs on public transport: differences in renderings on buses and minibuses.

**Law and official documents**

4. Chinese renderings of legal terms in Hong Kong and Singapore (for example, *tort*, *estoppel*).
5. Changes in the Chinese rendering of "reasonable person" in judgments across decades.
6. Company registration terms: 有限公司, 股份有限公司, and their English equivalents (Ltd., Inc.).
7. Bilingual consistency in Hong Kong legislation: how 須 / 可 correspond to *shall* / *may*. (e-Legislation is one possible source; see Section 3 on copyright.)

**Literature and culture**

8. Wuxia vocabulary in English translations of Jin Yong's novels, for example 輕功 and 內力 across translators.
9. Hong Kong and Taiwan Chinese translations of Murakami Haruki: titles and personal names.
10. English translation strategies for Cantonese song lyrics (for example, Beyond, Lam Jik).
11. Cultural localisation (festivals, food) in bilingual children's picture books.

**Media and subtitles**

12. Traditional Chinese subtitles for Korean and Japanese dramas: Hong Kong style versus Taiwan style for the same line.
13. Translation of Cantonese slang and profanity in Hong Kong drama subtitles: softened, replaced, or kept. (Profanity: follow the rules in Section 7.)
14. English renderings of Chinese news headlines, which often use exaggeration and ellipsis.
15. Bilingual subtitles on YouTube: user-generated versus professional subtitles.

**Business and products**

16. English labelling of nutrition and ingredient terms on food packaging (for example, 反式脂肪, 鈉).

17. Localisation of app interfaces: the same function named on iOS, Android, WeChat, and WhatsApp.
18. Traces of machine translation in e-commerce product descriptions: lexical features of human versus machine text.

**Language policy and sociolinguistics**

19. Consistency of Chinese and English terms across Hong Kong government websites (for example, 市民 versus 公民).
20. Chinese names of university faculties and schools (for example, *Faculty of Arts*) across universities.
21. Transliteration versus free translation of loanwords (咖啡, 沙律, 忌廉) in Hong Kong, Taiwan, and mainland China.
22. Lexical differences across the three regions (軟件/軟體, 地鐵/捷運, 視頻/影片).
23. Film and TV titles across Hong Kong and Taiwan: which lean toward free or commercial translation.
24. Changes in English renderings of policy terms over time (for example, 一國兩制).
25. Consistency and exceptions in MTR station names and public signage.

**Place names, menus, and proper nouns**

26. Translation of Hong Kong street names: colonial transliteration versus current romanisation, and free translation such as 花園道 *Garden Road*.
27. Translation of Cantonese dishes on menus: distribution of strategies (literal, transliteration, free, cultural compensation).
28. Multiple renditions of a term across domains (for example, *charge*, *consideration*, *bank* in law, finance, medicine, and IT).
29. English names of degrees and programmes (for example, *Bachelor of Arts*, *Master of Translation*) across universities.

**AI and machine translation (suited to this assignment)**

30. How machine translation handles Hong Kong place and personal names, compared with official renderings.
31. Differences in renderings of the same text by two or three LLMs.

## Appendix B. Topic checklist

1. Can I obtain 500 or more entries in one semester from public, traceable sources whose copyright status is acceptable?
2. Is there a visible translation pattern, strategy, or disagreement that I can classify?
3. Can I explain it with a theoretical framework (for example, cultural translation, skopos theory, norm theory)?

4. Will search, filters, and visualizations directly support my analysis?
5. If the topic involves profanity, political sensitivity, or personal data, do I have a plan for handling it?

## Appendix C. Working tips

- Test your annotation rules on 20 to 30 entries first, then expand to 500 or more.
- Record inter-annotator agreement if possible: re-annotate a sample after a break and calculate the agreement rate.
- Decide in advance which parts AI does and which parts you judge yourself, and record this in your AI Use Declaration.